Explainability as a non-functional requirement in an AI-assisted veterinary clinical system: a case study

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Abstract

Context. AI-assisted diagnostic modules in veterinary clinical-management systems frequently lack explainability operationalized as a verifiable requirement.
Objective. To determine which explainability requirements, operationalized as verifiable non-functional requirements (NFRs), are necessary and sufficient for the different user profiles (technical and non-technical) of an AI-assisted veterinary clinical system.
Method. Applied single-case study grounded in Requirements Engineering. Requirements were elicited through semi-structured interviews and direct observation, analyzed qualitatively, operationalized as functional and non-functional requirements, and linked through requirements traceability.
Results. Open coding of 16 semi-structured interviews yielded 167 open codes, consolidated into 50 axial codes across 7 categories; closed-coding saturation was reached (1.33 new axial codes on average in the final three interviews, 2.67% of the cumulative total, below the 5% threshold). Acceptance of AI-assisted diagnostic suggestions was consistently conditional: participants required the veterinarian's ability to review and correct suggestions (4/16) and grounding in validated literature and data-protection safeguards (6/16), with general moderate-to-high acceptance (11/16) and AI treated as decision support rather than a substitute for professional judgement. These conditions were operationalized as non-functional requirement RNF-18, traced to the AI-assisted diagnostic use case CU-06.
Conclusions. Explainability for AI-assisted veterinary diagnostic support is

best operationalized around contestability and provenance of suggestions rather than technical model interpretability alone, and can be linked, through requirements traceability, to concrete verifiable requirements. Broader validation across additional cases and a second independent coding round remain as future work.

Keywords: Explainability, Non-functional requirements, AI-assisted diagnosis, Veterinary informatics, User studies

1 Introduction

Artificial intelligence (AI)-assisted diagnostic support is increasingly embedded into clinical management systems, including veterinary practice, where automated recommendations directly influence treatment decisions taken by veterinary professionals and are also relevant to non-technical administrative staff who manage clinical workflows and reporting around those decisions. The SGCV-IA system — a real-world veterinary clinic management system developed for a client organization in the province of Los Ríos, Ecuador — incorporates an AI-based diagnostic module whose outputs must be trusted and actionable by users with markedly different technical backgrounds. Studying Requirements Engineering in this context matters because, in AI-assisted clinical settings, the absence of explicit and verifiable explainability requirements can directly undermine user trust and, ultimately, patient (animal) safety.

Despite growing attention to explainability in software engineering, most existing work operationalizes explainability as a general software quality attribute rather than as a set of verifiable, stakeholder-specific non-functional requirements (NFRs) elicited through empirical fieldwork – a persistent gap in industry-relevant empirical Requirements Engineering research more broadly [1]. Chazette et al. [2] propose a definition, model, and knowledge catalogue for explainability as a software quality concern, and Chazette et al. [3] extend this into an end-to-end framework spanning requirements analysis to system evaluation. However, empirical validation of this framework in real-world, resource-constrained, Latin American clinical domains — and specifically in veterinary informatics — remains scarce, leaving a gap in understanding how explainability requirements should be operationalized and validated for heterogeneous user profiles (technical vs. non-technical) in such settings.

This work contributes: (1) an empirical characterization of explainability concerns for a real-world AI-assisted veterinary diagnostic system; (2) their operationalization as explicit and verifiable non-functional requirements; and (3) traceability connecting stakeholder evidence, requirements, and system functionality.

Accordingly, this study addresses two research questions: **RQ1** asks what types of explanations users of the SGCV-IA system require to appropriately understand and assess the automated suggestions produced by the AI-assisted diagnostic component; **RQ2** asks to what extent these explainability concerns can be operationalized as explicit and verifiable non-functional requirements within the SGCV-IA requirements specification.

2 Related Work

The related work follows the abbreviated systematic-review scheme of Kitchenham and Charters [4], adapted to the scope of a course-based empirical study. The search string `("explainability" OR "explainable AI" OR "XAI") AND ("requirements engineering" OR "non-functional requirement") AND (healthcare OR clinical OR veterinary OR "human-centred")` was run against Google Scholar, the database accessible to the authors at the time of writing, on 5 September 2026, returning approximately 2,860 records; IEEE Xplore, ACM Digital Library, and Scopus were not queried directly due to institutional access constraints and are noted as a limitation in Section 6.2. The search covered publications from 2019 (the year in which explainability was first framed explicitly as a non-functional requirement by Köhl et al. [5]) through 2026.

Titles and abstracts were screened against the following inclusion criteria: (i) addresses explainability, interpretability, or trust as a software quality or requirements concern; (ii) addresses Requirements Engineering challenges specific to AI-based or machine-learning-based systems; or (iii) reports empirical evidence on explainability or AI-assisted decision support in healthcare, veterinary, or comparable human-centred clinical contexts. Exclusion criteria were: purely algorithmic or model-level explainability work with no requirements or software-engineering framing; studies not available in English or Spanish; and non-peer-reviewed sources other than arXiv preprints from established research groups in the area. The search was executed on 5 September 2026 and returned approximately 2,860 records; the first 30 records, ordered by relevance, were screened by title and abstract. Four records were excluded outright as likely non-peer-reviewed, low-quality outputs (single-author ResearchGate-only postings with near-identical titles on "adaptive learning for non-functional requirement classification," none indexed by a recognized venue), and one further record meeting the topical criteria was excluded for being a non-arXiv preprint (Research Square) not yet peer-reviewed, per the exclusion criteria above. Screening of the remaining records against these criteria yielded the 19 primary studies compared in Table 1, selected for direct relevance to the research questions and methodological framing of the present study.

Research on explainability in software engineering has increasingly moved beyond model-level interpretability toward the consideration of explainability as a property that must be addressed during Requirements Engineering. Köhl et al. [5] were among the first to frame explainability explicitly as a non-functional requirement. Chazette and Schneider [6] further characterize explainability as a non-functional requirement and discuss challenges associated with its elicitation and operationalization. Chazette et al. [2] provide a definition, model, and knowledge catalogue, while their subsequent work connects requirements analysis, design, and evaluation [3]. Deters et al. [7] develop a quality model for explainability, and Droste et al. [8] propose and evaluate an operational taxonomy of explanation needs. Building directly on that taxonomy, Obaidi et al. [9] compare interviews, focus groups, and online surveys as elicitation methods for explainability requirements in an industrial case, finding that interviews yielded the highest number of distinct explanation needs per participant – a methodological result consistent with the choice of semi-structured interviews as the primary elicitation technique in the present study. Together, these studies support treating explainability as an engineered and elicitable software concern rather than exclusively as a property of a machine-learning model.

A second strand concerns AI-based systems and the emergence of requirements challenges that differ from those of conventional software. Habiba et al. [10] discuss synergies between Requirements Engineering and Explainable AI in a position paper proposing a user-centric research agenda for explainability requirements. Ahmad et al. [11] identified 43 primary studies in a systematic mapping of RE4AI and reported limited empirical evaluation of ethics, explainability, and trust. A more recent systematic review of generative AI for Requirements Engineering [12] confirms that empirical, stakeholder-grounded studies remain a minority relative to tool-focused contributions. Habiba et al. [13] subsequently mapped 126 primary studies and identified requirements specification, explainability, and the gap between machine-learning engineers and end users among the prevalent RE4AI challenges. Habibullah et al. [14] show that machine-learning systems have quality demands that differ from traditional systems and examine how practitioners currently handle non-functional requirements. Heyn et al. [15] provide complementary empirical evidence from ten practitioner interviews concerning requirements for training data and runtime monitoring in critical ML systems. These studies indicate that AI-related requirements need to account for uncertainty, data, runtime behaviour, human stakeholders, and quality attributes beyond traditional functional behaviour.

A third strand focuses on users, healthcare, and the relationship between explanations and human judgement. Balasubramaniam et al. [16] connect transparency and explainability with requirements derived from ethical concerns, while de Brito Duarte et al. [17] examine explainability in relation to warranted trust. Freyer et al. [18] synthesize ethical reasons for explainability in healthcare AI decision-support systems, and Mienye et al. [19] review explainable AI in healthcare and its adoption challenges. Rosenbacke et al. [20] show that explanations can increase or decrease clinicians' trust depending on context. Habiba et al. [21] further report, from interviews with ML practitioners, diverse interpretations of explainability and recurring difficulties in communicating explainability needs to non-technical stakeholders. In a related vulnerable-population context, Xiao et al. [22] combine explainable machine learning on survey data with semi-structured interviews to identify which human aspects of older adults, caregivers, and developers drive requirements priorities in aged-care digital health, and report substantial misalignment between stakeholder groups – a finding methodologically and thematically close to the technical/non-technical stakeholder comparison pursued in the present study.

The literature also provides methodological support for making requirements explicit and verifiable. Großer et al. [23] empirically compare requirement-template systems and show that structured requirement formulations can affect quality characteristics such as ambiguity and readability. Umar and Lano [24] review automated support for Requirements Engineering and show that human intervention remains common across automated RE activities. Finkbeiner and Siber [25] address explainability requirements from a formal perspective by modelling them as hyperproperties, illustrating that explainability can be formulated as a system property that is subject to verification.

Veterinary informatics provides a comparatively less developed application domain for this line of research. Lustgarten et al. [26] describe veterinary informatics as a field connecting veterinary practice with computational and biomedical methods. Although veterinary AI research has expanded, the literature identified here does not provide a comparable body of empirical Requirements Engineering studies focused specifically on explainability requirements in veterinary clinical-management systems.

Taken together, the reviewed literature supports four observations. First, explainability is increasingly recognized as a software-engineering concern that should be addressed during Requirements Engineering rather than exclusively after model development [2, 3, 5–9]. Second, AI-based systems introduce requirements challenges involving explainability, trust, ethics, uncertainty, data, and the relationship between technical developers and end users [10, 11, 13–15]. Third, explainability is stakeholder- and context-dependent, and explanations do not uniformly produce appropriate trust [17, 18, 20–22]. Fourth, veterinary informatics remains comparatively underexplored for empirical Requirements Engineering studies focused on explainability [26].

The resulting research gap is therefore not simply the absence of veterinary research on artificial intelligence. Existing literature already provides conceptual frameworks, taxonomies, quality models, and healthcare evidence concerning explainability. The unresolved problem is the empirical integration of these perspectives into a Requirements Engineering process in which stakeholder evidence from a real veterinary organization is transformed into explicit and verifiable explainability non-functional requirements and traced to the corresponding system functionality.

The present study addresses this gap through the SGCV-IA case. In particular, it examines explainability requirements associated with the AI-assisted diagnostic functionality and their relationship with RNF-18 and use case CU-06. The study therefore contributes an empirical Requirements Engineering perspective on explainability in a veterinary clinical-management context.

3 Methodology

This study adopts an applied single-case design grounded in Requirements Engineering. The case is the SGCV-IA system, a veterinary clinical-management system incorporating an AI-assisted diagnostic functionality. The methodological focus of the present study is the elicitation, analysis, specification, and traceability of requirements, with particular attention to explainability as a non-functional requirement of the AI-assisted diagnostic component.

3.1 Research questions and PICOC

The study is guided by the following research questions:

- **RQ1.** What types of explanations do users of the SGCV-IA system require to appropriately understand and assess the automated suggestions produced by the AI-assisted diagnostic component?

Köhl et al. [5]	2019	Conceptual study	N/A	Conceptual framing	First frames explainability explicitly as an NFR	We operationalize an NFR from primary stakeholder data, not conceptually.
Chazette and Schneider [6]	2020	Conceptual study	N/A	Literature-grounded conceptualization	Establishes explainability as an NFR; identifies elicitation challenges	We report an empirical elicitation and operationalization in a real case.
Chazette et al. [2]	2021	Framework study	N/A	Definition, model, knowledge catalogue	Provides a conceptual model and dimensions of explainability	We map empirical stakeholder codes onto these dimensions in a specific domain.
Chazette et al. [3]	2022	Framework study	N/A	RE-to-evaluation framework	Connects requirements analysis, design, and evaluation	We instantiate this life cycle with one verifiable NFR traced to a real use case.
Ahmad et al. [11]	2023	Systematic mapping	43 primary studies (RE4AI corpus)	Systematic mapping	Limited empirical evaluation of ethics, explainability, trust in RE4AI	We contribute an empirical study addressing this gap in an unexplored domain.
Habibullah et al. [14]	2023	Empirical study	ML practitioners	Practitioner-focused empirical investigation	ML systems have NFR quality demands distinct from traditional systems	We isolate explainability specifically and trace it end-to-end via requirements traceability.
Balasubramaniam et al. [16]	2023	Conceptual study	N/A	Conceptual linkage	Connects ethical transparency concerns with requirements	Grounded in primary interview data rather than conceptual argument.
Freyer et al. [18]	2024	Systematic review	Healthcare AI decision-support literature	Secondary synthesis	Synthesizes ethical reasons for explainability in healthcare AI	Veterinary domain; primary empirical (interview) data rather than secondary synthesis.
Mienye et al. [19]	2024	Review	Healthcare XAI literature	Narrative review	Surveys XAI adoption challenges in healthcare	Primary empirical RE study, not a literature review.
Habiba et al. [13]	2024	Systematic mapping	126 primary studies (RE4AI corpus)	Systematic mapping	Identifies explainability and stakeholder gaps in RE4AI	Addresses one identified gap directly with primary stakeholder data.
Heyn et al. [15]	2024	Empirical interview study	10 ML practitioners	Semi-structured interviews	Challenges specifying data/runtime requirements for critical ML systems	Our interviewees are clinic end users, not ML engineers, and the focus is explainability.
Deters et al. [7]	2024	Framework study	N/A	Literature-review-grounded quality-model construction	Provides a quality model for specifying and evaluating explainability	We derive requirements directly from open/axial coding of interviews, not a predefined quality model.
Rosenbacke et al. [20]	2024	Systematic review	Clinician-facing XAI studies	Secondary synthesis	Explanation effects on clinician trust are context-dependent	Primary qualitative evidence from a specific (veterinary) clinical context.
Droste et al. [8]	2025	Empirical study	End users of everyday software systems	Taxonomy development and validation	Provides an operational taxonomy of explanation needs	Domain-specific application (AI-assisted veterinary diagnosis) rather than general software.
Obaidi et al. [9]	2025	Empirical comparison study	2 focus groups, 18 interviews, 188-respondent survey (industrial IT case)	Comparison of interviews, focus groups, and surveys as elicitation methods	Interviews yielded the highest number of distinct explanation needs per participant	Confirms semi-structured interviews as an effective elicitation method for explainability, used here in a clinical rather than industrial IT setting.
Habiba et al. [10]	2022	Position paper	N/A	Conceptual discussion of RE–XAI synergies	Proposes a user-centric research agenda linking RE practices to XAI challenges	We provide the empirical, single-case evidence this research agenda calls for.
Habiba et al. [21]	2025	Interview study	ML practitioners	Semi-structured interviews	Diverse interpretations of explainability; communication gaps with non-technical stakeholders	We interview the non-technical stakeholders directly, not practitioners' perceptions of them.
Sporsem et al. [27]	2026	Longitudinal case study	Development team and hospital clinicians (Norwegian hospital, paediatric AI system)	18-month longitudinal case study, iterative explanation design	Clinicians trusted the AI system when able to compare predictions against their own assessment ("evaluative requirements"), not when given technical/mechanistic explanations	Veterinary rather than human-paediatric setting; single-round retrospective case study rather than 18-month longitudinal fieldwork.
Xiao et al. [22]	2026	Mixed-methods study	103 older adults, 105 developers, 41 caregivers (aged-care digital health)	Explainable-ML analysis of survey data, validated with 12 semi-structured interviews	Identifies human aspects driving requirement priorities and misalignment between stakeholder groups	Different vulnerable-population domain (aged care vs. veterinary); we use interviews as the primary elicitation method rather than as a validation step for

- **RQ2.** To what extent can the identified explainability concerns be operationalized as explicit and verifiable non-functional requirements within the SGCV-IA requirements specification?

The research questions are structured according to PICOC:

- **Population (P):** stakeholders involved in the clinical and organizational context of the veterinary system, including veterinary professionals and other relevant users identified during requirements elicitation.
- **Intervention (I):** the elicitation and operationalization of explainability concerns as explicit non-functional requirements for the AI-assisted diagnostic functionality.
- **Comparison (C):** comparison of stakeholder perspectives and requirements evidence where differences or convergences are identified in the empirical material.
- **Outcome (O):** identified explainability concerns, their transformation into verifiable requirements, and their traceability to stakeholder evidence and system functionality.
- **Context (C):** the real-world organizational context associated with a veterinary clinic in Ecuador and the development of the SGCV-IA system.

3.2 Propositions

Following case study guidelines for software engineering [28], this study is guided by analytical propositions rather than statistical hypotheses, consistent with its qualitative, single-case design. Three propositions guided data collection and analysis:

- **P1.** Stakeholders of the SGCV-IA system will condition their acceptance of AI-assisted diagnostic suggestions on the ability to review, contextualize, or correct those suggestions, rather than accepting them unconditionally or rejecting AI-assisted diagnosis outright.
- **P2.** Explainability concerns raised during elicitation can be consolidated into a closed set of thematic categories that stabilizes (in the sense of closed-coding saturation, Section 4.1) within the 16 interviews collected for this case.
- **P3.** At least one explainability concern identified through open and axial coding can be operationalized as an explicit, traceable non-functional requirement linked to a specific use case and system component.

These propositions are evaluated against the empirical evidence in Section 4 and revisited in the Conclusions (Section 7).

3.3 Case study design

The study follows the case study approach described by Runeson and Höst [28]. SGCV-IA constitutes the single case under investigation. The analysis is embedded in the Requirements Engineering process and considers stakeholder evidence, requirements artefacts, and the resulting traceability relationships.

The case was investigated in the context of real veterinary clinical and administrative activities. The empirical material was collected during the requirements-elicitation stage and subsequently incorporated into the requirements specification. The study does not treat the AI component as an isolated machine-learning artefact; instead, the

AI-assisted diagnostic functionality is analyzed as part of a socio-technical software system in which professional users remain responsible for interpreting and acting upon the generated suggestions.

3.4 Participants and recruitment

The requirements-elicitation evidence identifies participants using anonymized identifiers P01–P16. The project repository maintains the correspondence between participant identifiers and real identities exclusively within a restricted area; the public project documentation does not reproduce this correspondence. This anonymization procedure is explicitly documented in the final ERS.

Participants were selected from the organizational context relevant to the veterinary system and contributed information concerning clinical workflows, administrative activities, information needs, and expectations regarding AI-assisted diagnostic support. The documented evidence includes interviews with veterinary professionals and observations of activities performed in the veterinary clinical setting.

Participation was voluntary and based on informed consent. Identifying information and original multimedia materials are handled separately from the anonymized research artefacts. The final ERS specifies that signed consent materials containing identifying information are stored in the restricted repository area, while anonymized or masked copies are used for dissemination.

3.5 Data collection

The principal data-collection techniques documented in the final ERS were semi-structured interviews and direct observation.

Semi-structured interviews. Interviews were conducted with veterinary professionals to identify current workflows, operational problems, information needs, expectations regarding the system, and conditions associated with the use of AI-assisted diagnostic suggestions. The interview instrument and corresponding evidence are included in the project repository. The documented evidence includes interview records associated with anonymized participant identifiers.

Direct observation. Observation was used to complement interview evidence by examining clinical and administrative activities in their organizational context. The final ERS records observation evidence concerning patient registration, clinical information management, and other processes relevant to the system requirements.

Documentary evidence. Organizational documents and other project artefacts were reviewed when necessary to understand processes and support requirements specification. These materials complement, rather than replace, the primary field evidence.

No questionnaire or survey is treated as a requirements-elicitation instrument in this stage of the study. The final ERS explicitly states that requirements information for this stage was obtained through semi-structured interviews and observation.

3.6 Requirements analysis

The collected evidence was analyzed to identify stakeholder needs and transform them into structured requirements. The process consisted of four main activities: (1) identification of relevant statements and needs in the empirical evidence; (2) abstraction and consolidation of recurring needs; (3) formulation of functional and non-functional requirements; and (4) traceability of the resulting requirements to their evidence and system artefacts.

Particular attention was given to explainability concerns associated with the AI-assisted diagnostic functionality. Explainability was treated as a software quality concern and subsequently operationalized as a non-functional requirement, following the Requirements Engineering perspective established by Chazette et al. [2, 3, 6].

In the final requirements specification, RNF-18 addresses explainability of the AI-assisted diagnostic functionality. The requirement is associated with the diagnostic use case CU-06 and specifies information that must be presented to the veterinary professional together with the AI-generated suggestion. This traceability connects the explainability concern to an identifiable requirement and system function.

3.7 Requirements traceability

Traceability was used to maintain the relationship between empirical evidence, raw requirements, formal requirements, system functionality, and design artefacts. The final ERS reports an extended traceability matrix containing 60 rows and links requirements with their corresponding sources, objectives, normative considerations, design elements, and associated artefacts.

This approach reduces the risk that explainability requirements become detached from the stakeholder evidence from which they originated. It also provides a basis for subsequent verification because each formal requirement can be related to an identifiable source and to the functionality in which it is intended to be realized.

3.8 Ethical and data-protection considerations

The study involves human participants and therefore incorporates informed consent and confidentiality measures. Participant identities are represented by anonymized identifiers P01–P16 in the public project materials. The mapping between these identifiers and real identities is retained exclusively in a restricted repository location.

Original consent documents, audio, video, and other potentially identifying materials are separated from the public research artefacts. The project documentation states that the handling of personal data follows the requirements established under Ecuador's Ley Orgánica de Protección de Datos Personales [29], in a manner methodologically comparable to model-based approaches for tracing data-protection legal requirements against system requirements developed for other regulatory regimes [30].

No personally identifying participant information is reproduced in this manuscript.

3.9 Data analysis and reproducibility

The qualitative analysis is supported by interview transcripts, observation evidence, documentary material, and thematic coding. The project repository contains anonymized transcripts, coding materials, requirements artefacts, and traceability information. The final ERS documents the organization of these materials and distinguishes restricted evidence from materials intended for dissemination.

The qualitative thematic-coding results derived from the 16 interviews are reported in Section 4. Any additional terminal quantitative or statistical validation (e.g., statistical tests, effect sizes, confidence intervals) is not reported in this manuscript version and will be incorporated only after the corresponding experimental evidence and analysis scripts have been completed and verified.

3.10 Deviations from the pre-registered protocol

Following the transparent-reporting practice recommended for pre-registered studies [31], deviations between the protocol registered in the Open Science Framework (OSF) during Entrega 3 (2A) and the analysis reported in this manuscript are documented here.

The registered protocol specified a two-round quantitative validation component for the AI-assisted diagnostic suggestion functionality (Round 1 and Round 2, six participants per round: three technical and three non-technical users), applied through a dedicated "Guion de validación v2.0" instrument built around a simulated diagnostic-suggestion stimulus, comprehension and satisfaction Likert scales, and a checklist of coverage against the four dimensions of the Chazette et al. explainability framework [2, 3, 6]. This instrument was fully designed and finalized but was not applied to participants within the project timeline. Consequently, the quantitative analyses planned for this component – coverage of the explainability framework (RQ2), Cohen's κ between validation rounds, and a Mann-Whitney U comparison of technical versus non-technical user perceptions – could not be completed for this version of the manuscript. This is a genuine deviation from the registered protocol rather than a planned staging of results, and it is treated as a limitation of the present study (Section 6.4) and as explicit future work (Section 7). The qualitative findings reported in Section 4 (RQ1) were instead derived from the broader, already-completed elicitation interviews (P01–P16), following the same explainability framework for interpretation.

3.11 Methodological limitations

The single-case design provides detailed contextual evidence but does not by itself establish broad generalizability across veterinary organizations or AI-assisted clinical systems. The study therefore emphasizes contextualized requirements evidence and traceability rather than population-level statistical inference. The implications of this design for internal, external, construct, and conclusion validity are discussed in Section 6.

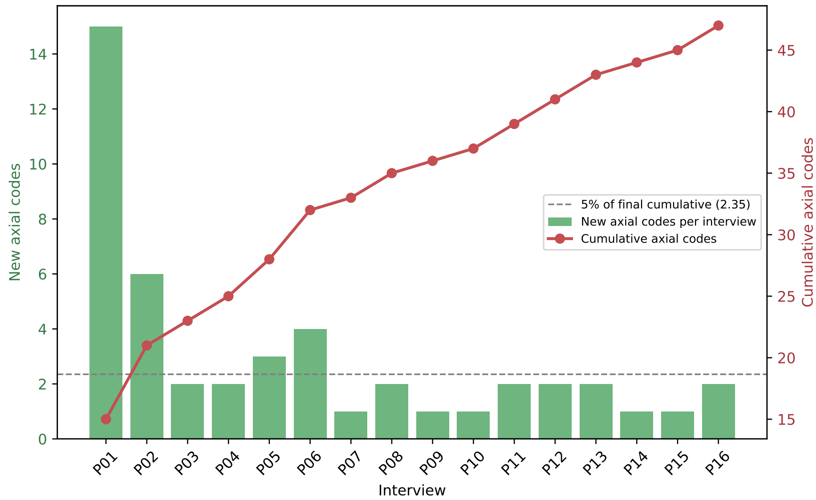

Fig. 1 Closed (axial) thematic saturation curve for the SGCV-IA interview corpus (n=16, 50 axial codes). Bars show new axial codes contributed per interview (left axis); the line shows the cumulative axial code count (right axis). The open-coding curve (167 raw codes, not saturated by the same criterion) is reported as process evidence in the replication package.

4 Results

4.1 Thematic saturation

Open coding was applied line-by-line to the 16 anonymized interview transcripts (P01–P16), yielding 167 open codes. These were consolidated through axial coding into 50 codes across 7 categories: clinical records and patient management, administrative and financial management, inventory, appointments and communication with owners, perception of artificial intelligence, usability and performance, and security. Coding followed the collection order (P01→P16) to track the rate at which additional interviews contributed new codes, separately at the open-coding level and at the level of the closed (axial) coding scheme, following the code-saturation approach of Hennink et al. [32].

At the open-coding level, new codes continued to appear throughout the corpus (16 new codes at P01, falling to a range of 7–16 per subsequent interview, with 10, 11, and 12 new open codes in the final three interviews, P14–P16 respectively); this pattern is expected, since open coding preserves each participant's own wording before consolidation and is not itself the unit on which saturation is assessed. At the level of the closed axial scheme, new-category emergence declined sharply: the first two interviews contributed 15 and 6 new axial codes respectively, and the final three interviews (P14–P16) contributed only 1, 1, and 2 new axial codes, an average of 1.33 new codes, corresponding to 2.67% of the final cumulative total of 50 — below the 5% threshold adopted for closed-coding saturation. Figure 1 reports both curves. The 16-interview sample therefore supports closed-coding saturation of the categorical scheme, while individual open-coded expressions of these categories continued to vary in wording up to the final interview.

Table 2 Axial codes associated with explainability, trust, and adoption conditions for the AI-assisted diagnostic functionality (category: perception of artificial intelligence)

Code	Description	Freq. (n=16)
IA-1	Moderate-to-high acceptance of or confidence in AI-assisted diagnostic suggestions (self-reported agreement/trust)	11/16
IA-4	Requires conditions such as validated scientific literature, data protection, or periodic updates to trust AI suggestions	6/16
IA-2	Rejects or limits automated/remote diagnosis without physical evaluation of the patient	5/16
IA-3	Requires that the veterinarian review/correct AI suggestions before applying them	4/16
IA-5	Skepticism toward unvalidated internet sources (e.g., search engines, chatbots) consulted by pet owners	1/16
IA-6	Interest in AI reducing workload for record-keeping and treatment tasks	1/16
IA-7	AI-assisted dose calculation by patient weight	1/16
IA-8	AI-assisted messaging/chatbot for appointment scheduling	1/16

4.2 Explanation needs and conditions for trust (RQ1)

The category "Perception of artificial intelligence" (8 axial codes, 33 underlying open codes) concentrates the codes most directly relevant to RQ1. Table 2 summarizes these codes, relabelled IA-1 to IA-8 for readability, together with their frequency across the 16 participants (a participant is counted once per code regardless of how many open-coded fragments of theirs mapped to it). Two findings stand out. First, acceptance of AI-generated diagnostic suggestions was consistently conditional rather than unconditional: eleven of sixteen participants (IA-1, 11/16) expressed moderate-to-high agreement or confidence in AI-assisted diagnostic suggestions, but four participants (IA-3, 4/16) explicitly required that the veterinary professional retain the ability to review and correct every suggestion before it is acted upon, and six participants (IA-4, 6/16) conditioned their trust on the recommendation being grounded in validated scientific literature, data protection, or comparable safeguards. Second, five participants (IA-2, 5/16) explicitly rejected or limited remote/automated diagnosis in the absence of physical evaluation of the patient, reinforcing that AI support was framed as complementary rather than substitutive of in-person clinical judgement.

These conditions (IA-3, IA-4) map onto the explainability-as-quality-concern perspective established by Chazette et al. [2, 3, 6]: the requirement to review and correct suggestions (IA-3) corresponds to the need for actionable, contestable explanations, and grounding in validated literature together with data-protection guarantees (IA-4) corresponds to trustworthiness and privacy-related explanation content. Code IA-2 reinforces this pattern from a different angle: participants consistently preserved in-person clinical judgement as the point at which an AI-generated suggestion must be confirmed, indicating that explanations are expected to support, and not replace, professional discretion. A single, closely related code in the clinical-records category (participant P13's request to attach photographic evidence of wound or lesion evolution to the patient record) is discussed separately in Section 4.4 as a low-frequency emergent finding rather than as part of the core explainability pattern.

4.3 Operationalization as a verifiable requirement (RQ2)

The explainability concerns identified in Section 4.2 were operationalized in the final requirements specification as RNF-18 (Section 3.6), traced to the AI-assisted diagnostic use case CU-06. The information content required by RNF-18 – namely, that the AI-generated suggestion must be accompanied by information enabling the veterinary professional to review, contextualize, and correct it – was derived directly from codes IA-3 and IA-4. This traceability link connects a specific, auditable non-functional requirement to the empirical stakeholder evidence from which it originated, illustrating one instance of the elicitation-to-specification operationalization sought by RQ2. A complete requirement-by-requirement coverage matrix against the full 47-code scheme remains part of ongoing work (Section 7).

4.4 Low-frequency and emergent codes

Several low-frequency codes are reported as negative or atypical cases rather than omitted, following standard qualitative-analysis practice. Skepticism toward unvalidated internet sources such as search engines or chatbots used by pet owners (IA-5, 1/16) and interest in AI as a means of reducing administrative and record-keeping workload (IA-6, 1/16) indicate that the perception of AI extended beyond the core diagnostic-trust pattern reported in Section 4.2, without displacing it. Outside the perception-of-AI category, the request for photographic documentation of clinical evolution (wounds, lesions) linked to the patient record, raised by a single participant (P13), was among the last new axial codes to appear in the corpus and is noted here as a candidate requirement for future iterations rather than as a validated finding.

5 Discussion

5.1 Implications for research

The findings extend prior explainability frameworks [2, 3, 6] to a domain, veterinary informatics, in which comparable empirical Requirements Engineering evidence was previously scarce [26]. The conditional-acceptance pattern observed here (IA-1, IA-2, IA-3) is consistent with the context-dependent trust effects reported by Rosenbacke et al. [20] in human healthcare settings, suggesting that the relationship between explanation content and clinician trust may generalize across at least some high-stakes, professionally mediated AI-assisted domains. The explicit mapping between empirical codes and the explainability dimensions proposed by Chazette et al. (Section 4.2) also provides a concrete, replicable procedure for connecting qualitative stakeholder evidence to an existing explainability framework, addressing part of the empirical-integration gap identified in Section 2.

5.2 Implications for practice

For practitioners designing AI-assisted clinical-support functionality, the results suggest that explainability requirements should prioritize contestability (the ability to review and correct a suggestion, IA-3) and provenance (grounding in validated literature and data protection, IA-4) over purely technical measures of model interpretability. The consistent restriction of AI-informed decisions to physical, in-person evaluation by the veterinary professional (IA-2) further suggests that role-based access and accountability controls should be treated as a companion requirement to explainability rather than as a separate concern, since participants did not distinguish clearly between "being able to understand a suggestion" and "having the authority to act on it."

5.3 Surprising or counterintuitive findings

Two observations were not anticipated at the outset of the study. First, outright rejection of AI-assisted diagnosis was markedly less frequent than conditional acceptance (IA-1, 11/16, against IA-2, 5/16, most of which qualifies rather than rejects AI support), suggesting that resistance to AI-assisted diagnostic tools in this context is better characterized as a demand for adequate safeguards than as opposition to the technology itself. Second, the request for photographic documentation of clinical evolution emerged only in interview P13, close to but not exactly at the point of closed-coding saturation (P14–P16), illustrating that even a corpus that satisfies a closed-coding saturation criterion can surface a low-frequency but potentially actionable requirement late in the elicitation process at the open-coding level; this reinforces the value of reporting both the open- and closed-coding saturation curves (Section 4.1) rather than relying on a single aggregate criterion.

6 Threats to Validity

The validity of this study is discussed according to the four categories commonly used in empirical software engineering: internal, external, construct, and conclusion validity [33, 34]. The discussion is adapted to the characteristics of a qualitative, single-case Requirements Engineering study and complements the case-study guidance of Runeson and Höst [28].

6.1 Internal validity

A first threat concerns researcher interpretation during the transformation of interview and observation evidence into formal requirements. Because the study involves qualitative analysis, the identification, abstraction, and consolidation of stakeholder needs may be influenced by the analysts' judgement. This threat was mitigated by maintaining traceability between empirical evidence, raw requirements, formal requirements, use cases, and design artefacts. The extended traceability matrix provides an auditable chain linking requirements to their documented sources. Nevertheless, interpretive decisions cannot be eliminated completely, and alternative analysts could classify or formulate some stakeholder concerns differently.

A second threat arises from the possibility that interview statements alone may not fully represent actual clinical and administrative practice. To reduce dependence on a single source of evidence, semi-structured interviews were complemented with direct observation and documentary evidence from the organizational context. This triangulation strengthens the consistency of the elicited requirements. However, the observations were conducted within the same organizational context as the interviews, so they do not provide an independent replication of the findings in another veterinary organization.

6.2 External validity

The main threat to external validity is the single-case design. SGCV-IA was investigated within a specific veterinary organizational context in Ecuador, and the stakeholders, workflows, constraints, and regulatory environment may differ from those of other veterinary clinics, countries, or AI-assisted clinical systems. The study therefore does not claim statistical generalization. Instead, the case, stakeholder roles, elicitation procedures, requirements artefacts, and traceability relationships are documented in detail to support analytical comparison with similar contexts, as recommended for software-engineering case studies [28]. The residual limitation is that transferability to other organizations must be assessed by future studies in additional settings.

A second, narrower threat concerns the literature search reported in Section 2: it was executed against Google Scholar rather than the full set of databases recommended for systematic searches in software engineering (Scopus, IEEE Xplore, ACM Digital Library), due to institutional access constraints at the time of writing. This may have excluded relevant venue-indexed studies not well covered by Google Scholar's ranking, and the exact number of records returned by the documented search string was not independently re-verified by a second reviewer. The screening criteria and search string are nonetheless reported in full to allow replication with a broader database set in future work.

A second threat concerns domain specificity. Explainability needs identified for an AI-assisted veterinary diagnostic functionality may not transfer directly to human healthcare, other high-stakes domains, or AI systems used for different tasks. The study mitigates this threat by grounding its contribution in Requirements Engineering concepts and by explicitly identifying the SGCV-IA context and the diagnostic use case to which RNF-18 is traced. Even so, empirical replication is required before broader claims can be made about explainability requirements across domains.

6.3 Construct validity

A primary construct-validity threat is the operationalization of explainability. Explainability is a broad and context-dependent quality concern, and reducing it to a finite set of software requirements may omit dimensions relevant to particular stakeholders. To mitigate this threat, explainability was treated explicitly as a non-functional requirement following established Requirements Engineering work on explainability [2, 3, 6]. In SGCV-IA, the concern is operationalized through RNF-18 and traced to the AI-assisted diagnostic use case CU-06, providing a concrete connection between the abstract construct and system behaviour. The residual limitation is that one operationalization cannot be assumed to cover every possible interpretation of explainability.

A second threat is the potential conflation of explainability with related constructs such as transparency, trust, usability, or diagnostic accuracy. The present study reduces this risk by focusing the analysis on the information that must accompany an AI-generated suggestion and on its specification as a verifiable software requirement, rather than treating user trust or model performance as equivalent measures of explainability. Nevertheless, stakeholder statements may naturally combine these concerns, and qualitative coding cannot completely isolate them from one another.

6.4 Conclusion validity

The principal threat to conclusion validity at the present stage is the qualitative and contextual nature of the available evidence. The conclusions that can be supported by this part of the study concern the identification, specification, and traceability of explainability requirements; they do not establish population-level effects or statistically significant differences between user groups. This threat is mitigated by restricting claims to the documented evidence and by preserving traceability from stakeholder material to the resulting requirements. Accordingly, no statistical inference is reported in this manuscript version.

A second threat concerns the incompleteness of the terminal empirical validation at the time of this manuscript version: as documented in Section 3.10, the two-round quantitative validation component registered in the OSF protocol was designed but not executed within the project timeline. Statistical tests, effect sizes, confidence intervals, and conclusions dependent on that dataset are consequently absent from this manuscript version, rather than staged for a later section. This prevents unsupported quantitative conclusions, but it also means that the present study cannot yet make claims about the measured effectiveness, satisfaction, or comparative performance of the operationalized explainability requirements. Those claims remain outside the scope of the current manuscript version and are carried forward as future work (Section 7).

7 Conclusions and Future Work

With respect to **RQ1** (what types of explanations users require), the thematic analysis of 16 interviews shows that veterinary professionals and staff require AI-generated diagnostic suggestions to be reviewable and correctable, grounded in validated scientific literature, and accompanied by guarantees of patient-data protection, while non-clinical staff are explicitly excluded from acting on such suggestions without professional oversight. With respect to **RQ2** (the extent to which these concerns can be operationalized as verifiable non-functional requirements), the study shows that at least one central explainability concern can be translated into an explicit, traceable requirement: RNF-18, linked to use case CU-06 and to the empirical codes that motivated it (IA-3, IA-4).

Taken together, these findings contribute (i) an empirical account of explainability and trust conditions for AI-assisted diagnostic support in a veterinary clinical-management context, a domain in which comparable empirical Requirements

Engineering evidence was previously scarce; (ii) a concrete mapping between qualitative stakeholder evidence and an established explainability framework [2, 3, 6]; and (iii) a traceable example of operationalizing an explainability concern as a verifiable non-functional requirement within a real requirements specification.

Future work includes: (1) executing the two-round quantitative validation component registered in the OSF protocol (Section 3.10) – comprehension and satisfaction scales, Chazette framework coverage, Cohen's κ between rounds, and the technical/non-technical Mann-Whitney comparison – which was designed but not applied within the present project timeline; (2) completing a second independent coding round to compute inter-coder reliability (Fleiss' κ [35]), as noted in the project's thematic-coding documentation; (3) extending the requirement-to-code coverage matrix beyond RNF-18 to systematically trace all remaining explainability-relevant codes (Table 2) to corresponding functional and non-functional requirements; and (4) validating the resulting explainability requirements with additional veterinary organizations to assess transferability beyond the single-case design used in this study.

Data and materials availability

The materials supporting the requirements-elicitation and analysis reported in this manuscript are maintained in the SGCV-IA project repository and in the accompanying anonymized replication package, deposited following the FAIR guiding principles for scientific data management [36, 37]. The available research artefacts include anonymized interview transcripts, thematic-coding materials (open and axial codes, with saturation curves and their generating scripts), requirements artefacts, and traceability information. Original consent forms and multimedia materials containing potentially identifying information are retained in a restricted repository area and are not publicly released. Anonymized or masked copies are used for dissemination where appropriate. The anonymized replication package is deposited on Zenodo at https://doi.org/10.5281/zenodo.22558095; a persistent Software Heritage identifier for the source repository [38, 39] and machine-readable citation metadata following the Citation File Format [40, 41] are provided in the repository's `CITATION.cff` file. Access to restricted materials is subject to the applicable ethical and data-protection conditions.

Use of AI-assisted technologies

During preparation and revision of this manuscript, an AI-assisted large language model (Claude, Anthropic) was used interactively, with human oversight, exclusively as a support tool for verification and language improvement. Its use was limited to: (i) checking the consistency, clarity, and wording of text previously prepared by the authors; (ii) verifying bibliographic and other factual data against the information and sources used by the authors; and (iii) suggesting improvements to grammar, spelling, style, clarity, and overall redaction. The AI tool was not used to generate original or substantive content, including research data, thematic codes, categories, analysis, results, methodology, interpretations, conclusions, tables, figures, or research decisions. All suggestions were reviewed by the authors before inclusion, and the authors remain fully responsible for the accuracy, provenance, and scientific integrity of all content and for the final version of the manuscript. No personal, confidential, or identifying participant information was included in prompts.

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